

4 Nanotechnology

4.1 Introduction

Nanotechnology is the term given to those areas of science and engineering where phenomena that take place at dimensions in the nanometre scale are utilised in the design, characterisation, production and application of materials, structures, devices and systems. Although in the natural world there are many examples of structures that exist with nanometre dimensions (hereafter referred to as the nanoscale), including essential molecules within the human body and components of foods, and although many technologies have incidentally involved nanoscale structures for many years, it has only been in the last quarter of a century that it has been possible to actively and intentionally modify molecules and structures within this size range. It is this control at the nanometre scale that distinguishes nanotechnology from other areas of technology.

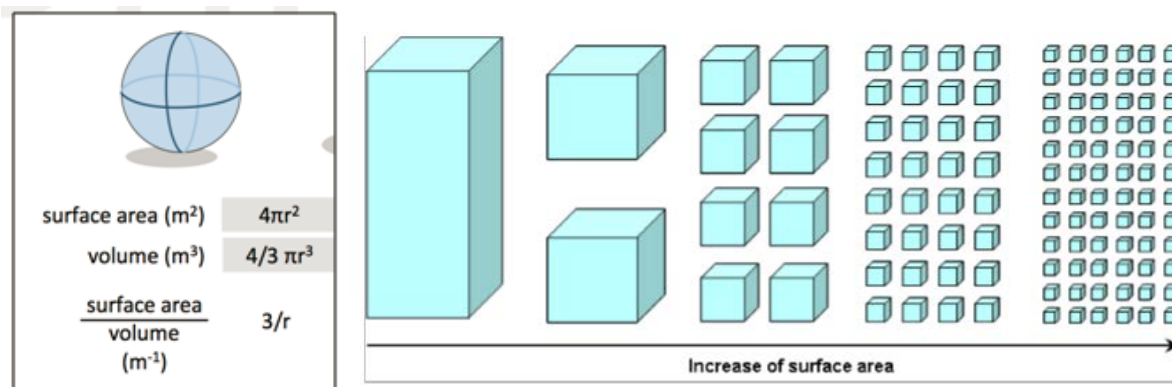
4.2 Why are nanomaterials different?

Nanomaterials due to their size regime behave in a different way as compared to their bulk counterparts. There are various reasons for the difference in behaviour. Here we discuss some of the major reasons for their difference in behaviour.

4.2.1 Surface-to-volume ratio:

As the size of materials approaches nanoscale, the percentage of atoms at the surface of a material becomes significant. Nanoparticles have a very high surface area to volume ratio.

Nanomaterials possess large surface areas. This can be understood as follows: when the bulk material is repeatedly divided into smaller and smaller pieces, total volume, of all the pieces taken together, does not change while the surface area considerably increases. Thus, the surface-area to volume ratio is greatly enhanced.

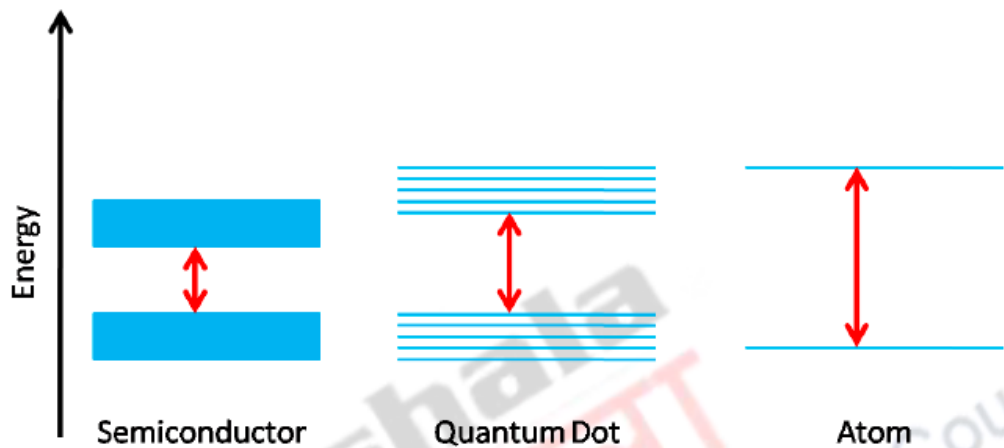


Atoms or molecules existing at the interface (or surface) differ from the same atoms/molecules present in the interior or bulk of the material. This applies to all materials. Surface atoms or molecules have increased reactivities and are highly prone to form clusters. Thus, the surface atoms are unstable and possess higher surface energies.

4.2.2 Quantum Confinement:

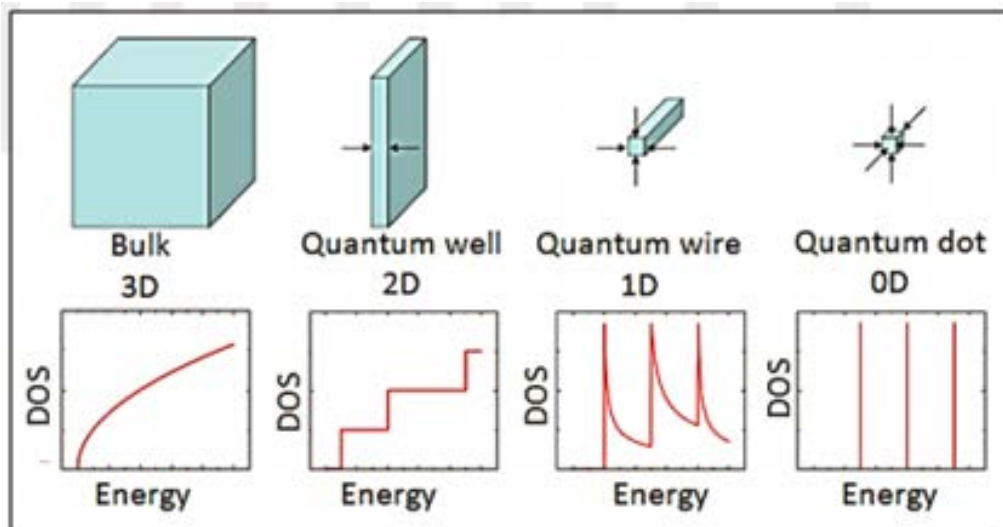
A bulk material or solid has three dimensions. If only one length of three-dimensional nanostructure is in nanoscale (1-100 nm), the structure is known as a quantum well. Instead, if two sides are of nanoscale (1-100 nm), the resulting structure is known as quantum wire. A nanoparticle is often referred to as a quantum dot as its all three dimensions are in the nanoscale (1-100 nm).

Quantum confinement results in increase in the energy bandgap of the material. Additionally, at nanoscale dimensions as the energy levels become quantized. Thus, the band overlap in metals disappears and a bandgap emerges. Thus, some metals behave as semiconductors at nanoscale.



4.2.3 Density of States:

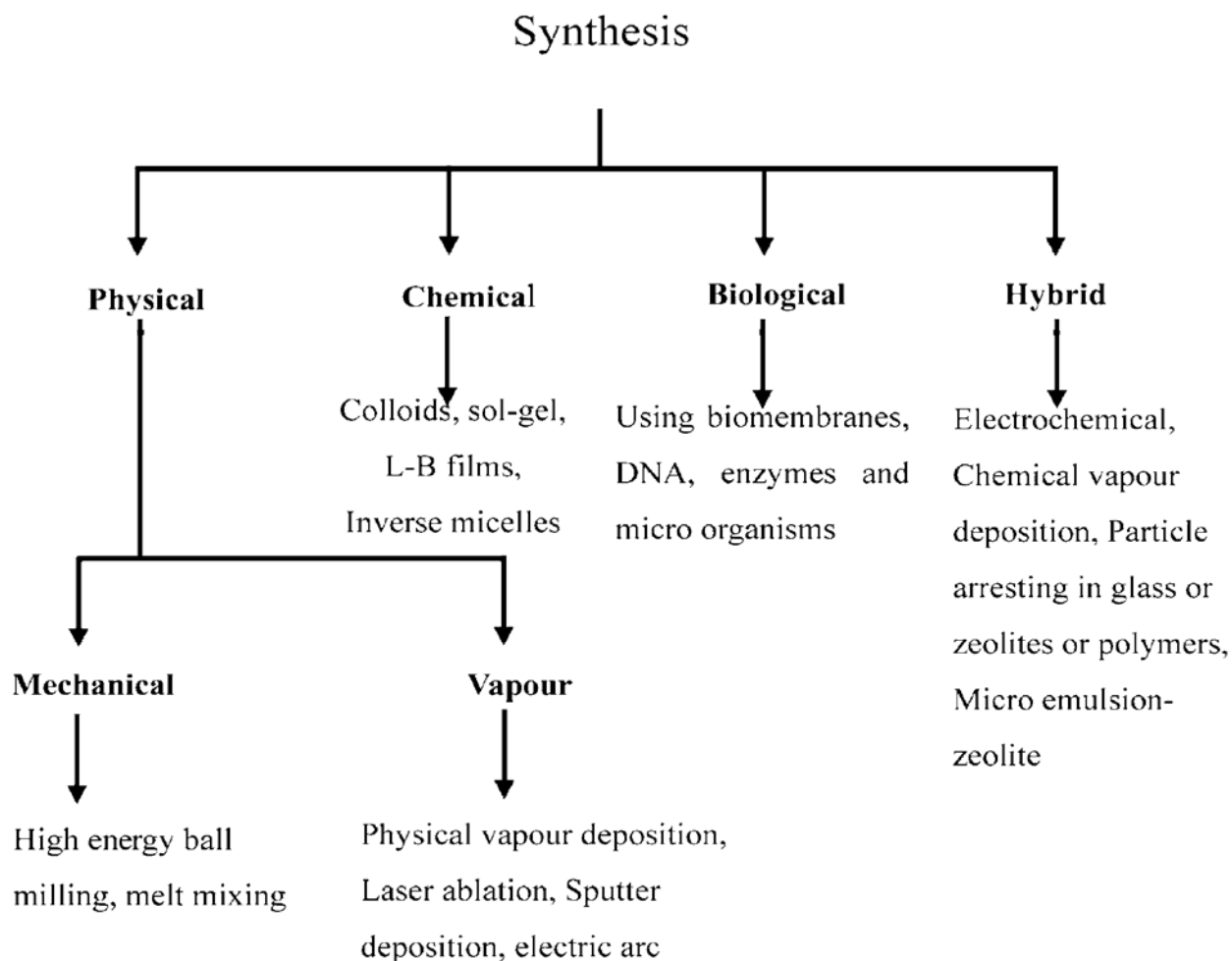
Density of states $D(E)$ is an important quantity. It is defined as the number of states per unit energy range available for occupancy of an electron.



As a result of quantization in states the electron distribution changes as we go from bulk to nano scale. Thus, resulting in changes the properties of the nanomaterials.

4.3 Synthesis of Nanomaterials:

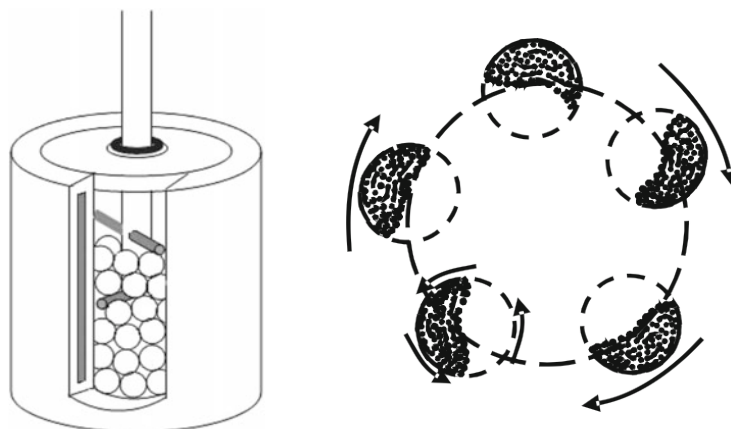
There are a large number of techniques available to synthesize different types of nanomaterials in the form of colloids, clusters, powders, tubes, rods, wires, thin films etc. Some of the already existing conventional techniques to synthesize different types of materials are optimized to get novel nanomaterials and some new techniques are developed. Nanotechnology is an interdisciplinary subject. There are therefore various physical, chemical, biological and hybrid techniques available to synthesize nanomaterials.



The technique to be used depends upon the material of interest, type of nanomaterial viz. zero dimensional (0-D), one dimensional (1-D) or two dimensional (2-D), their sizes and quantity.

4.3.1 High Energy Ball Milling:

It is one of the simplest ways of making nanoparticles of some metals and alloys in the form of powder. There are many types of mills such as planetary, vibratory, rod, tumbler etc. Usually, one or more containers are used at a time to make large quantities of fine particles. Hardened steel or tungsten carbide balls are put in containers along with powder



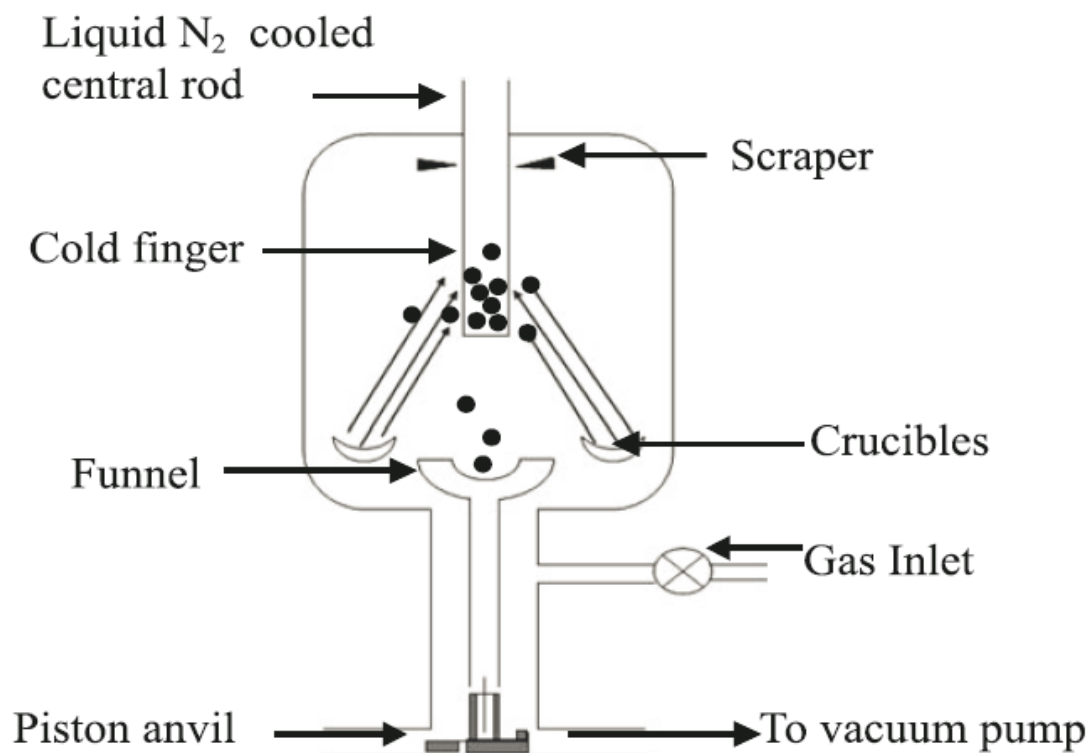
or flakes ($<50\text{ }\mu\text{m}$) of a material of interest.

Container is closed with tight lids. Usually 2:1 mass ratio of balls to material is advisable. If the container is more than half filled, the efficiency of milling is reduced. Heavy milling balls increase the impact energy on collision. Larger balls used for milling produce smaller grain size but larger defects in the particles. The process, however, may add some impurities from balls. A temperature rise in the range of $100\text{--}1,100\text{ }^{\circ}\text{C}$ is expected to take place during the collisions. Lower temperatures favour amorphous particle formation. During the milling, liquids also can be used. The containers are rotated at high speed (a few hundreds of rpm) around their own axis. Additionally, they may rotate around some central axis and are therefore called as 'planetary ball mill'. When the containers are rotating around the central axis as well as their own axis, the material is forced to the walls and is pressed against the walls as illustrated in Fig. By controlling the speed of rotation of the central axis and container as well as duration of milling it is possible to ground the material to fine powder (few nm to few tens of nm) whose size can be quite uniform.

Some of the materials like Co, Cr, W, Ni-Ti, Al-Fe and Ag-Fe are made nanocrystalline using ball mill. Few milligrams to several kilograms of nanoparticles can be synthesized in a short time of a few minutes to a few hours.

4.3.2 Physical Vapour Deposition:

This technique basically involves use of materials of interest as sources of evaporation, an inert gas or reactive gas for collisions with material vapour, a cold finger on which clusters or nanoparticles can condense, a scraper to scrape the nanoparticles and piston-anvil (an arrangement in which nanoparticle powder can be compacted). All the processes are carried out in a vacuum chamber so that the desired purity of the end product can be obtained.

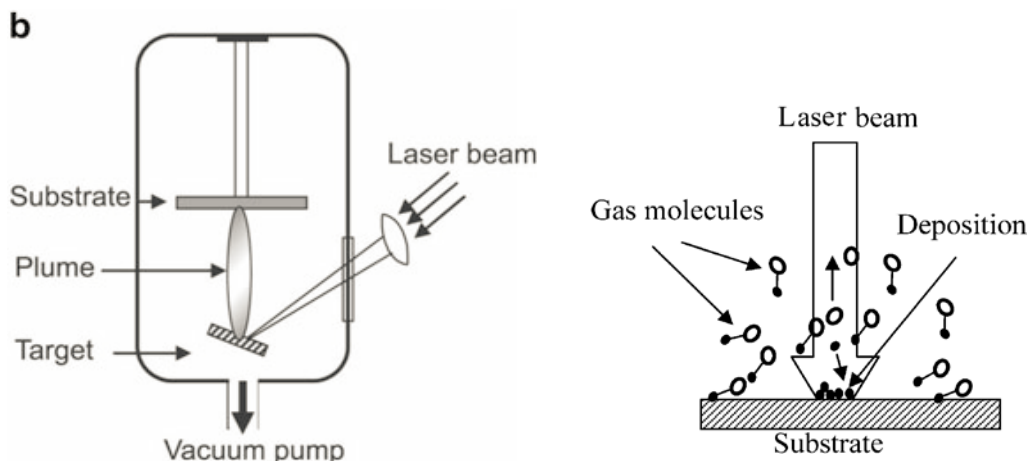


Usually, metals or high vapour pressure metal oxides are evaporated or sublimated from filaments or boats of refractory metals like W, Ta and Mo in which the materials to be evaporated are held. The density of the evaporated material close to the source is quite high and particle size is small (<5 nm). Such particles would prefer to acquire a stable lower surface energy state. Due to small particle or cluster interaction bigger particles get formed. Therefore, they should be removed away as fast as possible from the source. This is done by forcing an inert gas near the source, which removes the particles from the vicinity of the source. In general, the rate of evaporation and the pressure of gases inside the chamber determine the particle size and their distribution. Distance of the source from the cold finger is also important. Evaporated atoms and clusters tend to collide with gas molecules and make bigger particles, which condense on cold finger. While moving away from the source to cold finger the clusters grow. If clusters have been formed on inert gas molecules or atoms, on reaching the cold finger, gas atoms or molecules may leave the particles there and then escape to the gas phase. If reactive gases like O₂, H₂ and NH₃ are used in the system, evaporated material can interact with these gases forming oxide, nitride or hydride particles. Alternatively, one can first make metal nanoparticles and later make appropriate post-treatment to achieve desired metal compound etc. Size, shape and even the phase of the evaporated material can depend upon the gas pressure in deposition chamber.

4.3.3 Laser Vapourization (Ablation):

In this method, vapourization of the material is effected using pulses of laser beam of high power. The set up along with the interaction with the evaporation source is depicted in Fig.

The setup is an Ultra High Vacuum (UHV) or high vacuum system equipped with inert or reactive gas introduction facility, laser beam, solid target and cooled substrate. Clusters of any material of which solid target can be made are possible to synthesize. Usually laser operating in the UV range such as excimer (excited monomers) laser is necessary because other wavelengths like IR or visible are often reflected by surfaces of some metals.



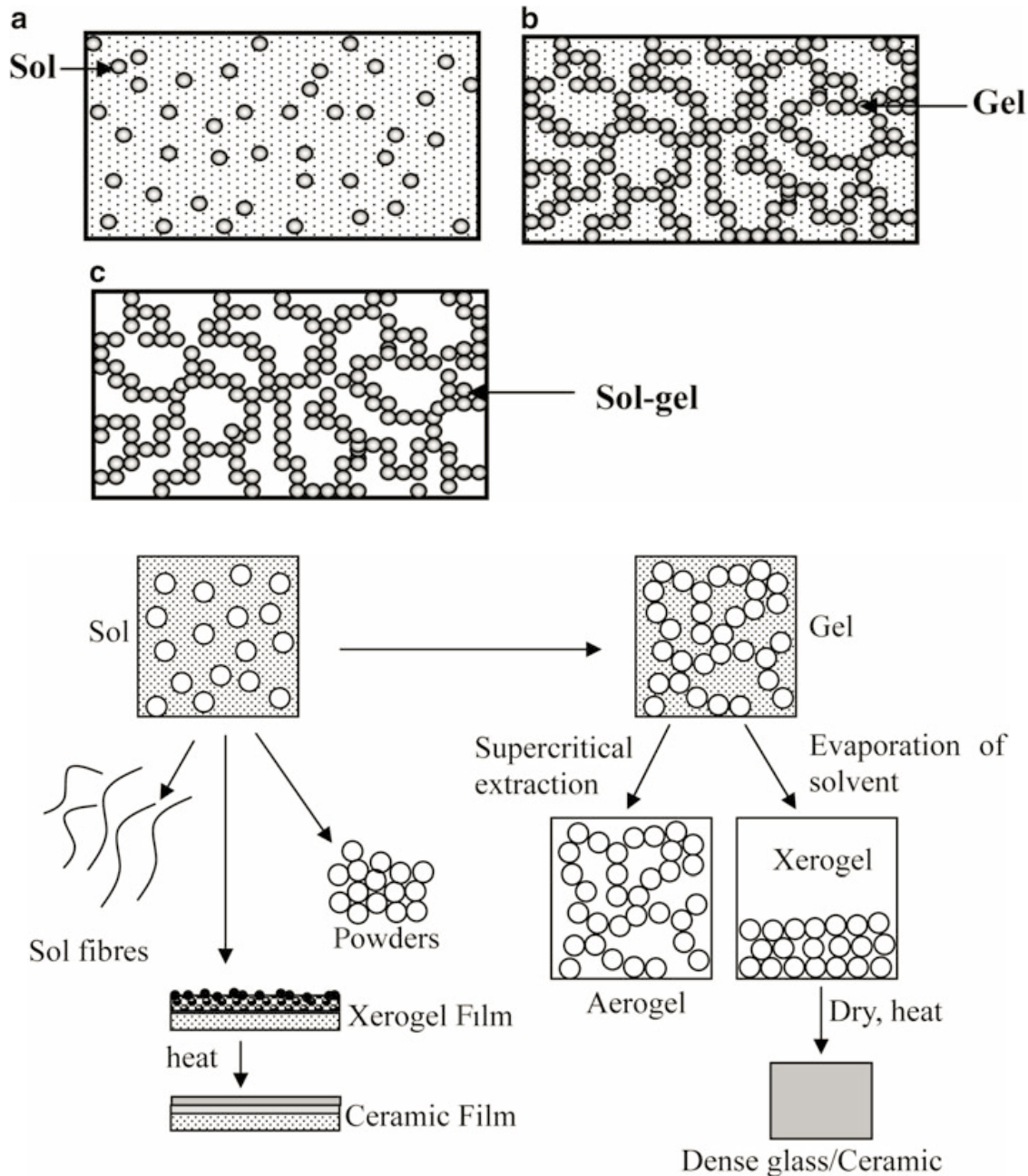
A powerful beam of laser evaporates the atoms from a solid source and atoms collide with inert gas atoms (or reactive gases) and cool on them forming clusters. They condense on the cooled substrate. The method is often known as laser ablation. Gas pressure is very critical in determining the particle size and distribution. Simultaneous evaporation of another material and mixing the two evaporated materials in inert gas leads to the formation of alloys or compounds. This method can produce some novel phases of the materials which are not normally formed. For example, Single Wall Carbon Nanotubes (SWNT) are mostly synthesized by this method.

4.3.4 Sol-Gel Method:

As the name suggests sol gel involves two types of materials or components, 'sol' and 'gel'. First of all sol gel formation is usually a low temperature process. This means less energy consumption and less pollution too. It is therefore not surprising that in the nuclear fuel synthesis it is a desired process. Although sol-gel process generates highly pure, well controlled ceramics it competes with other processes like CVD or metalloorganic vapours derived ceramics. The choice of course depends upon the product of interest, its size, instrumentation available and ease of processing. In some cases solgel can be an economical route, provided precursors are not very expensive. Some of the benefits like getting unique materials such as aerogels, zeolites, and ordered porous solids by organic-inorganic hybridization are unique to sol-gel process. It is also possible to synthesize nanoparticles, nanorods or nanotubes using sol-gel technique.

Sols are solid particles in a liquid. They are thus a subclass of colloids. Gels are nothing but a continuous network of particles with pores filled with liquid (or polymers containing liquid). A sol gel process involves formation of 'sols' in a liquid and then connecting the

sol particles (or some subunits capable of forming a porous network) to form a network. By evaporating the liquid, it is possible to obtain powders, thin films or even monolithic solid. Sol gel method is particularly useful to synthesize ceramics or metal oxides although sulphides, borides and nitrides also are possible.



Synthesis of sol-gel in general involves hydrolysis of precursors, condensation followed by polycondensation to form particles, gelation and drying process by various routes as shown in Fig. Precursors (starting chemicals) are to be chosen so that they have a tendency to form gels. Both alkoxides or metal salts can be used.

Although it is not mandatory that only oxides be formed by a sol-gel process, often oxide ceramics are best synthesized by a sol-gel route. For example in silica, SiO_4 group with Si

at the centre and four oxygen atoms at the apexes of tetrahedron are very ideal for forming sols with interconnectivity through the corners of tetrahedrons, creating some cavities or pores. By polycondensation process (i.e., many hydrolyzed units coming together by removal of some atoms from small molecules like OH), sols are nucleated and ultimately sol-gel is formed.

4.4 Properties of Nanomaterials:

Many properties of solids such as optical, electrical, magnetic, etc. depend on its size. As compared to its bulk form, properties of materials change when at least one of the dimensions of material is reduced to nanoscale (1-100 nm).

4.4.1 Optical Properties:

a) Optical properties of metallic nanoparticles

The color of nanoparticles is different from bulk material. One of the daily life examples is of glass. Glasses are transparent, but when doped with different nanoparticles, adopt different colors like red, pink, blue, green, etc. depending upon the dissolved nanoparticles.

Metal	Color for bulk	Color at nano-level
Gold	Yellow	Bright red
Silver	Colorless	Pale yellow

G. Mie explained the phenomenon using Maxwell's equation in 1908. When electromagnetic radiation incidents on spherical particles of uniform size a part of the radiation is absorbed and a part is scattered. Thus, the intensity of transmitted light changes.

- If incident light has wavelength λ , intensity I_0 , length of medium x , then

The intensity of transmitted radiation is given by:

$$I = I_0 e^{-\mu x}$$

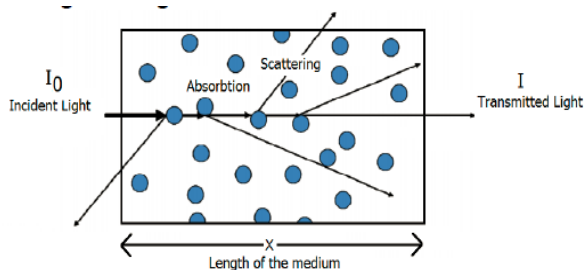
Where,

$\mu = \frac{N}{V} C_{ext}$ is extinction coefficient

N is number of particles in medium,

V- Volume of colloidal particles

C_{ext} – extinction cross section of a particle



At nanoscale, as size of the nanoparticles change, there is change in the length of medium and thus the values of μ and x changes.

This changes the intensity of transmitted light and it depends on the wavelength of incident light. This changes the color of nanoparticles at different nanoscale region

Mie theory successfully explained absorption of light of visible wavelength for nanoparticles, but for particle size less than ~ 10 nm, there is need to consider dielectric constant which depends also on particle size.

b) Optical properties of semiconductor nanoparticles

Every material has a characteristic size only below which size dependent properties are realized. In semiconductors this size is the size of the exciton

In semiconductor nanoparticles, as the particles become smaller and smaller, the energy gap increases and absorption peak shifts towards higher energy/frequency values

This leads to shift in the absorption spectrum and change in color of particles

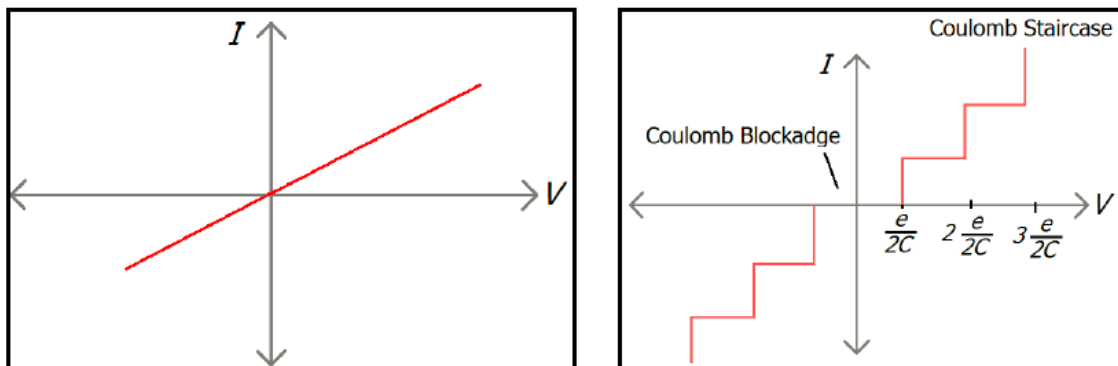
4.4.2 Electrical Properties:

Electrical properties are characterized by the mean free path of electron (path of the electron collision due to atoms or ions). The mean free path changes at nanoscale due to collision and tunneling within the conductor.

a) I-V Characteristics of bulk conductor:

When voltage V is applied across the conductor, current I flowing through it is governed by Ohm's Law and has a linear variation

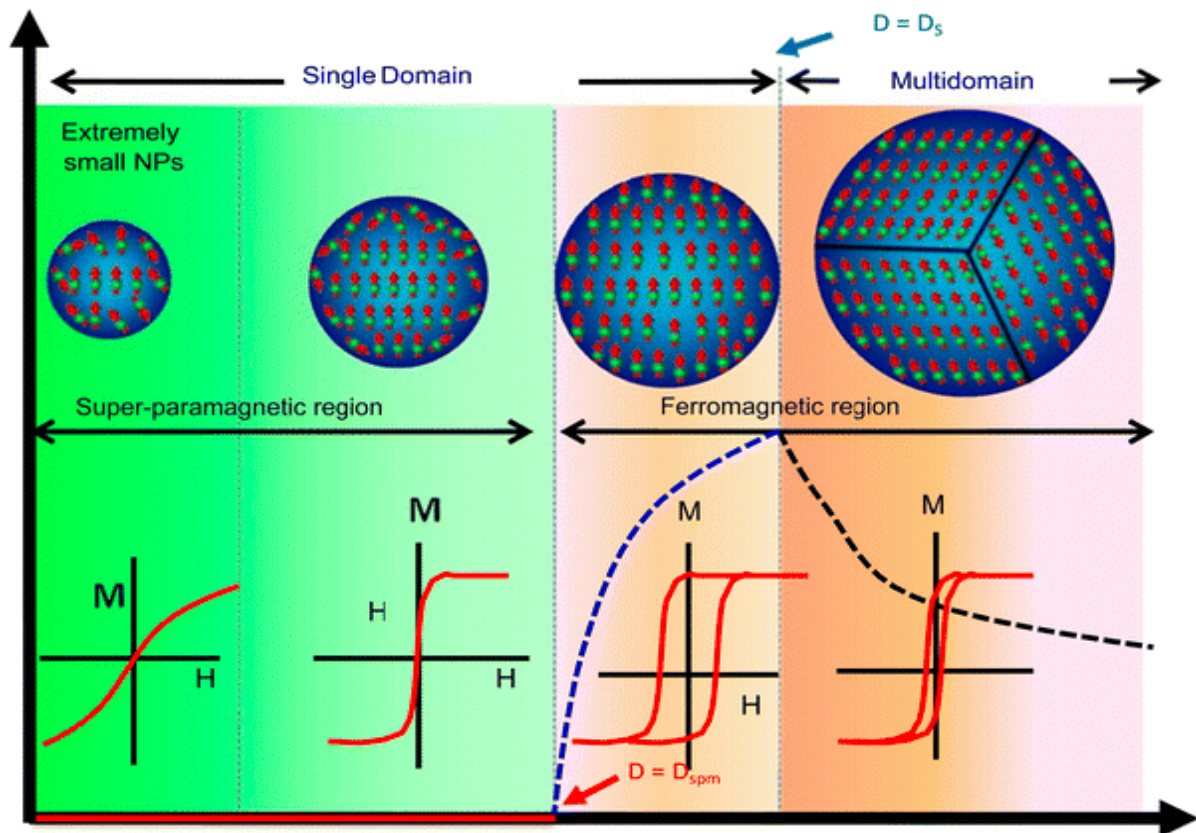
b) I-V Characteristics at nanoscale: If dimensions of conductor are in nanoscale (1-100 nm), there appears a region around zero voltage for which there is no current. A single electron is transferred by tunneling when the voltage is $\pm e/2C$. Therefore, when the voltage is less than this electron cannot be transferred. This gives a region of zero current at low bias voltage and is known as 'Coulomb blockade region'. The electrons are transferred when the voltage is $\pm e/2C$. The repeated tunneling of single electron produces 'Coulomb staircase'.



Electrical resistivity: The materials of nano-sized grains have larger number of grain boundaries than corresponding polycrystalline materials. Electrical resistivity of materials having nano-sized grain is larger than the polycrystalline materials due to scattering of electrons at grain boundaries.

4.4.3 Magnetic Properties:

Magnetic properties of the material can also be altered by nanostructuring. The magnetization curve of a material can be adjusted by controlling its size, resulting in soft or hard magnets with enhanced properties. Generally, the magnetic behavior of a material



depends upon its structure and temperature. For experiencing a magnetic field, the material must have a finite nonzero spin. The transition size of classically expected magnetic domains is $\sim 1\mu\text{m}$. With decreasing the size of the magnet, surface atoms form a significant portion of the total number of atoms. This greatly enhances surface effects, and in turn the quantum confinement assumes significance. As the size of these domains is reduced to nanoscale, the materials exhibit novel properties because of quantum confinement, like the giant Magneto-Resistance effect (GMR). This is a fundamental nano-effect which is now being used in modern data storage devices.

4.5 Applications of Nanomaterials:

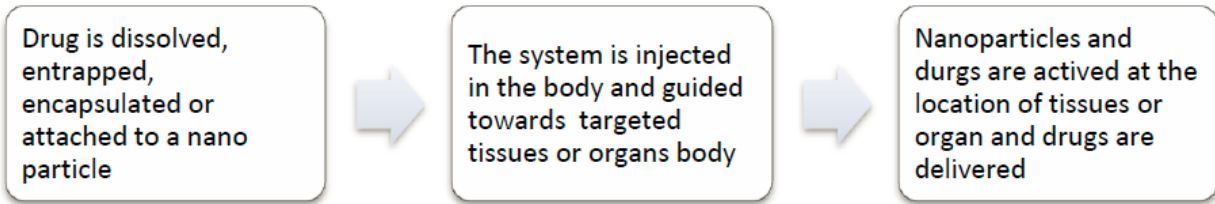
4.5.1 Medical Field:

Targeted Drug Delivery:

The traditional drug delivery system, the drug is distributed to the affected tissues or organs of the patient's body as well as to the healthy tissues or organs. This leads to the side effects.

In the targeted drug delivery system, the drug is dissolved, entrapped, encapsulated or attached to a nano particle. The system is then embedded in a capsule which is guided towards the affected part of the body

The capsule is opening at the specific tissues or organs controlled by externally applied magnetic field or infrared light or physiologically. Then drug can be delivered in controlled manner



Carriers for targeting drug:

Depending on the treatment required and nature of the drug, following are few examples of carriers are used for targeted drug delivery:

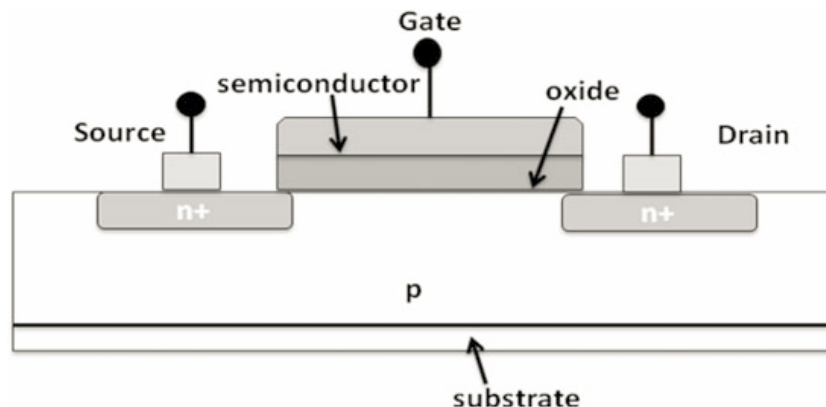
Sr.	Carriers of drug	Applications
1	Poly (alkyl cyanoacrylate) nanoparticles with anti cancer agents	Cancer therapy
2	Poly (methyl methacrylate) nanoparticles with vaccines	Vaccine delivery
3	Poly(alkyl cyanoacrylate) polyester nanoparticles with anti	Intracellular targeting
4	DNA- gelatin nanoparticles	DNA delivery

4.5.2 Electronics Industry:

In the spin-electronic devices, the spin property of electrons (instead of charge) is taken into consideration. Advantage with spin is that it cannot be easily destroyed by scattering from collisions with other charges, impurities or defects. The electronic devices with typical dimensions of few nanometers in either of three directions display unique properties. Such devices are fast, compact, and relatively cheap. Such devices are typically known as spin-electronic (spin-electronic) devices.

4.5.2.1 Single Electron Transistor:

Just as in a usual or classical micro size transistor (see Box 10.2) there are three main components of a single electron transistor viz. source, drain and gate. As illustrated in Fig., the quantum dot is placed between the source and the drain. The gate controls the raising or lowering of the electron levels in the quantum dot through the gate capacitor



by adding or lowering the number of electrons in the quantum dot. As the phenomenon of tunnelling and, hence, Coulomb blockade are the main phenomenon involved, the electrons are controlled precisely one by one and the name to the transistor is given as Single Electron Transistor. Different SET's are fabricated based on mainly the quantum dot' being used. The requirement is that there has to be nanostructure with discrete energy levels as in 'particle in a box' which takes place of the quantum dot so that a single electron control is achieved.

4.5.2.2 Spintronic devices:

The electronic devices with typical dimensions of few nanometers in either of three directions display not just the miniaturization but unique properties not known over last 5–6 decades since the beginning of solid-state devices. Single Electron Transistor (SET), spin valves, and Magnetic Tunnel Junctions (MTJ) are conceptually new devices based on nanotechnology. Such devices are fast, compact, relatively cheap and finding their way to market. Spin valve type devices are already being used in personal computers to 'read' disk which have enabled to increase data storage capacity of hard disks. Interestingly, spin valve and MTJ are based on a concept which itself is growing into an area in itself known as spintronics or spin-based electronics or magneto-electronics.

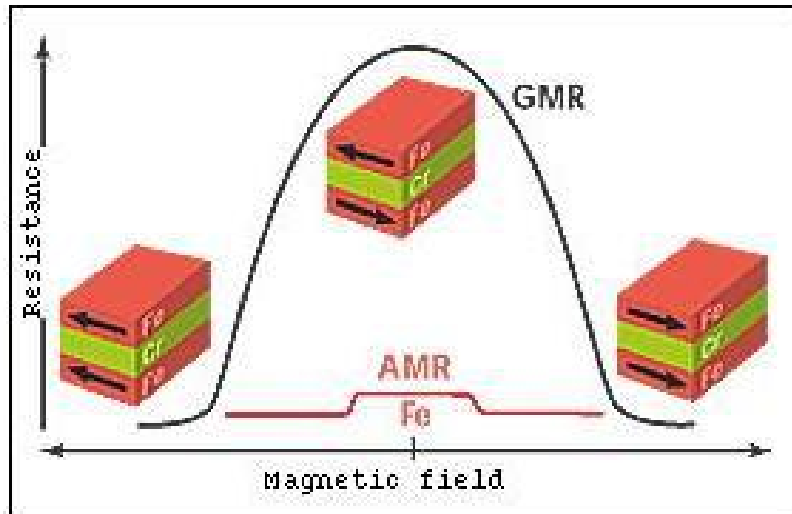
It is well understood that an electron (or hole) has both charge and spin. However, electronics has so far used only the charge property of electron (or hole) and spin has been neglected. It has been now realized in recent years that if spin of an electron (or hole) is taken into account, properly fabricated devices would lead to some superior devices. Using an external magnetic field, spin transport can be controlled. Advantage with spin is that it cannot be easily destroyed by scattering from collisions with other charges, impurities or defects. Many spin-based devices like Spin-FET, Spin-LED, Spin-RTD, optical switches with THz frequency, modulators, encoders, decoders, and q-bits for quantum computers are on the hot list of scientists and the technologists.

Some of the materials which have a potential in spintronics are summarized below.

1. II-VI semiconductors doped with transition metal ions (also known as Diluted Metal Semiconductors or DMS for short)
2. III-V semiconductors doped with transition metal ions (DMS)
3. Metal oxides with large band gap and doped with transition metals. For example TiO_2 , SnO_2 doped with cobalt
4. Eu chalcogenides
5. Heusler alloys like NiMnSb , Mn_2CoGe
6. Ferromagnetic metal oxides like CrO_2 , Fe_3O_4
7. $\text{Mn}_{11}\text{Ge}_8$
8. Lanthanum doped CaB_6

Giant Magneto-Resistance (GMR) devices:

Giant Magneto Resistance (GMR) can be realized in the multilayers. Deposition of one kind of material over the other (sputtering, e-beam evaporation or electrochemical deposition are commonly employed techniques) of a few nanometer thickness, and repeating it several times gives rise to a multilayer. Multilayers are artificially created man-made materials. Such multilayers are characterized by the presence of a large number of interfaces. The properties of multilayers are, therefore, governed not only by the parent materials but also by their surface and interface properties.



Magnetic multilayers in which ferromagnetic layers of materials few nanometers thick are separated by comparable thickness metallic layers, in these multilayers ferromagnetic layers can be ferromagnetically or antiferromagnetically coupled. This gives rise to a magnetoresistivity which depends upon the orientation of the magnetic layers. Magnetoresistance (MR) is the relative change in electrical resistance of a material on the application of magnetic field.

4.5.2.3 Nanophotonic devices:

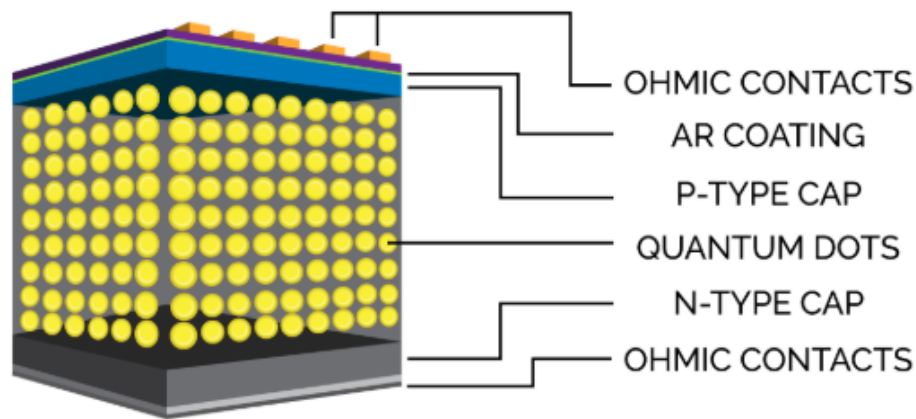
When nanostructures (quantum dots, nanowires or 2-D thin films) or nanocomposites are used to produce light or detect light we are already dealing with a branch in nanoscience known as Nanophotonics. It is expected that one day we will have nanophotonic chips just like semiconductor chips in which light production, propagation, manipulation like amplification, filter, detection etc. can be performed on a 'nanochip'. Manipulation of signals would then be faster.

We know that in a semiconductor, between the valence and the conduction band, there is an energy gap in which no electron states exist and is known as forbidden gap. It means that the electrons with energy between valence and conduction band are not allowed to propagate in the crystal. Similarly in a photonic band gap material, the photons of certain wavelength (or frequency) are forbidden to propagate. Thus, a photonic band gap results. Such a gap can be created artificially by arranging nanoparticles of small uniform size in

periodic lattice. The variation of size or dielectric constant is sensitive to optical gap and can be used to sense small variations in the photonic crystals.

4.5.2.3.1 Quantum Dot Solar Cell:

A quantum dot solar cell (QDSC) is a solar cell design that uses quantum dots as the captivating photovoltaic material. It attempts to replace bulk materials such as silicon, copper indium gallium selenide (CIGS) or cadmium telluride (CdTe). Quantum dots have bandgaps that are adjustable across a wide range of energy levels by changing their size. In bulk materials, the bandgap is fixed by the choice of material(s). This property makes quantum dots attractive for multi-junction solar cells, where a variety of materials are used to improve efficiency by harvesting multiple portions of the solar spectrum.



Principle: QD solar cells are based on a thin film of inorganic nanoparticles with dimensions smaller than their Bohr radius, resulting in quantum confinement of carriers. As a consequence of the quantum confinement, the band gap of QDs can be tuned and optimized for various device applications.

Quantum dots are essentially semi conducting particles whose size has been reduced even less than that of Exciton Bohr radius. As per the dynamics of quantum mechanics, the energy inside the electron tends to become finite unlike those present in the atoms. This is another reason as to why quantum dots are also referred to as artificial atoms. Whenever the sizes of these dots are altered, it ends up shaping and altering the energy levels and therefore the band gap gets defined accordingly. As the band gap can be tuned, these dots become best suitable for solar cells.

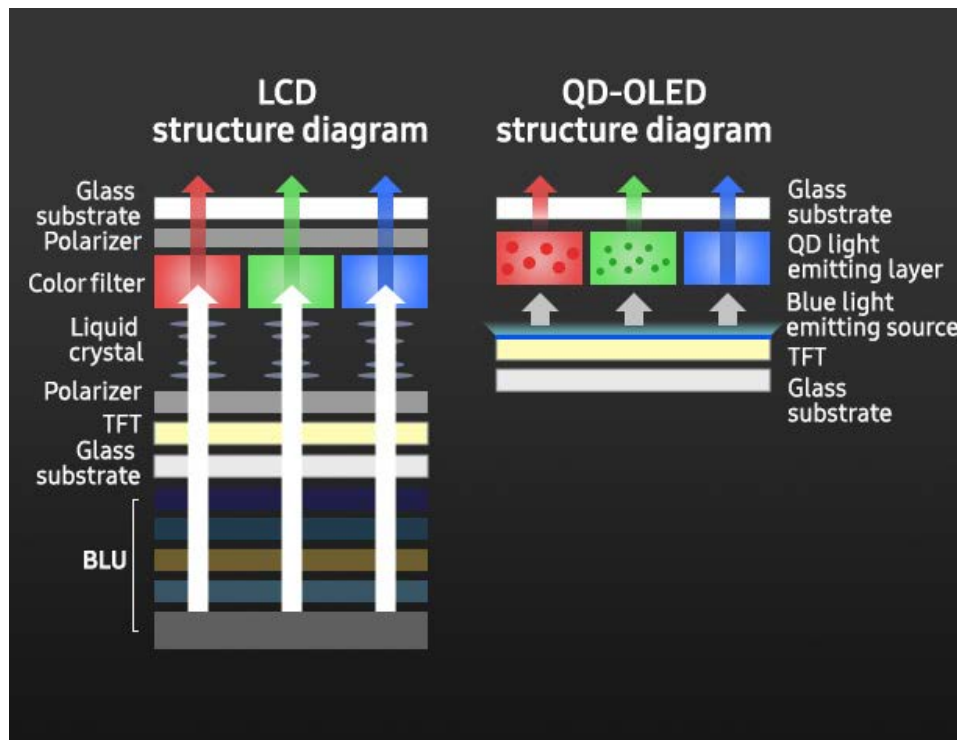
4.5.2.3.2 Quantum Dot LEDs:

A quantum dot display is a display device that uses quantum dots (QD), semiconductor nanocrystals which can produce pure monochromatic[a] red, green, and blue light. Photo-emissive quantum dot particles are used in LCD backlights or display color filters. Quantum dots are excited by the blue light from the display panel to emit pure basic colors, which reduces light losses and color crosstalk in color filters, improving display brightness and color gamut. Light travels through QD layer film and traditional RGB filters made from color pigments, or through QD filters with red/green QD color

converters and blue passthrough. Although the QD color filter technology is primarily used in LED-backlit LCDs, it is applicable to other display technologies which use color filters, such as blue/UV active-matrix organic light-emitting diode (AMOLED) or QNED/MicroLED display panels. LED-backlit LCDs are the main application of photo-emissive quantum dots.

Principle: A layer of quantum dots is sandwiched between layers of electron-transporting and hole-transporting organic materials. An applied electric field causes electrons and holes to move into the quantum dot layer, where they are captured in the quantum dot and recombine, emitting photons.

QDs are either photo-emissive (photoluminescent) or electro-emissive (electroluminescent) allowing them to be readily incorporated into new emissive display architectures. Quantum dots naturally produce monochromatic light, so they are more efficient than white light sources when color filtered and allow more saturated colors that reach nearly 100%.



QD-LEDs are characterized by pure and saturated emission colors with narrow bandwidth, with FWHM (full width at half maximum) in the range of 20–40 nm. Their emission wavelength is easily tuned by changing the size of the quantum dots. Moreover, QD-LED offer high color purity and durability combined with the efficiency, flexibility, and low processing cost of comparable organic light-emitting devices. QD-LED structure can be tuned over the entire visible wavelength range from 460 nm (blue) to 650 nm (red) (the human eye can detect light from 380 to 750 nm). The emission wavelengths have been continuously extended to UV and NIR range by tailoring the chemical composition of the QDs and device structure.

4.5.3 Automobile Industry

Nanotechnology has a wide range of applications in automobiles. Nanomaterials in various forms are used for designing sturdy structural parts, smooth and non-scratch paints, self-cleaning windows, better tires, reduction of harmful emission, etc. A few of them are listed below:

1. Sturdy structural parts

Various body parts of vehicles are made up of steel, some alloys, rubbers, plastics etc. Nanotube composites have mechanical strength better than steel

2. Smooth and non-scratch Paints

Cars are spray painted with fine particles. Nanoparticle paints provide smooth, non-scratch thin attractive coating

3. Self-cleaning glass for windows

Self-cleaning glass can be made by dissolving small amount of titania (TiO_2) nanoparticles while manufacturing glass. Titania is able to dissociate organic dust in presence of UV light available in the sunlight. Drops of water on glass give hazy look, but TiO_2 -containing glass can spread water evenly giving clear sight.

4. Small motor parts

Small motors are needed in vehicle (such as wipers, window glass movements, etc). Very powerful electric motors are made using shape memory alloys using nanoparticles of materials like Ni–Ti. They require less power and gives better performance.

5. Better tires

Tires of vehicles undergo wear and tear with use and also increase its weight. By using nanoparticle clay, better, light weight, less rubber consuming thinner tires are possible that can reduce the overall weight, increase in fuel consumption and speed.

6. Controlling harmful emission

Use of efficient nanomaterial catalysts is one solution to convert harmful emission into less harmful gases. Large surface area of nanoparticles is useful to produce better catalysts. They are capable of absorbing emission of particles and poisonous gases like CO and NO from vehicle exhausts.

7. Hydrogen fuel

Instead of using conventional petro fuels, there are numerous advantages of using hydrogen as a fuel. When hydrogen fuel is burned it can only produce harmless water vapor. Hydrogen gas is normally stored in a metal cylinder under high pressure. Nanocylinders of carbon nanotubes have potential to store hydrogen in very small space.